



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO D.K.R FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

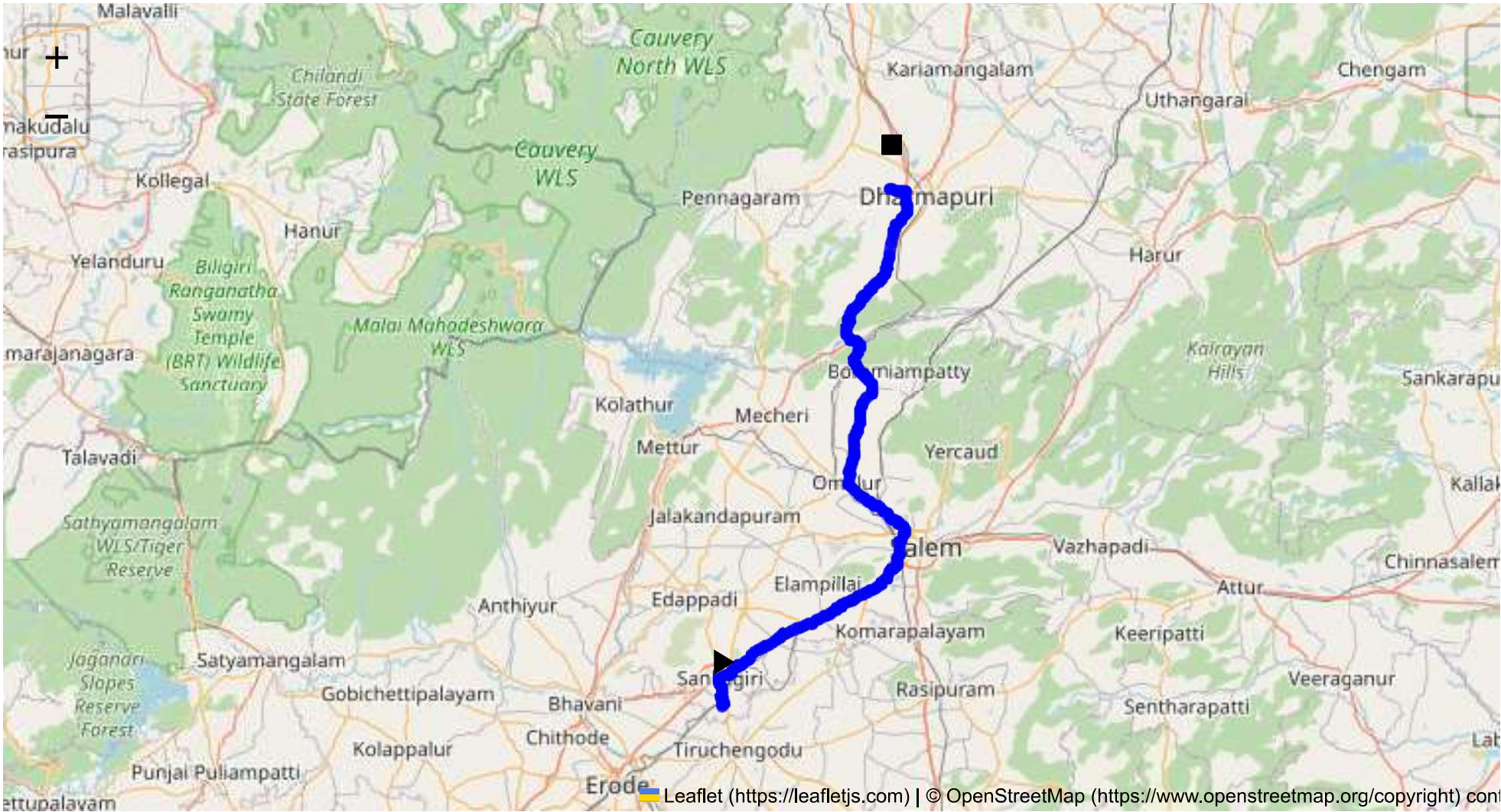
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 107.72 km
Estimated Duration: 2.1 hours
Adjusted Duration (Heavy Vehicle): 2.6 hours
Start: (11.4381, 77.8734)
End: (12.143924, 78.108592)



Welcome to the Journey Risk Management Study

Route Safety Analysis: From CVQF+23W, Sangagiri to 44V5+HC Pappinaickanahalli, Tamil Nadu

1. Overview of the Route Map:

The route from Sangagiri to Pappinaickanahalli covers approximately 107.72 kilometers, passing through several small towns and rural areas in Tamil Nadu. The typical driving duration for heavy vehicles is around 2.07 hours, with a mix of state highways and local roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- **Climate:** Tamil Nadu has a tropical climate, characterized by hot summers, moderate-to-heavy monsoons, and mild winters.
- **Weather Hazards:**
 - **Monsoon Season (June to September):** Heavy rains can cause flooding and landslides, especially in rural areas, affecting road conditions.
 - **Heat (April to June):** High temperatures can lead to road surface deterioration and heat stress for drivers.

3. Analysis of Traffic Patterns:

- **Peak Traffic Hours:** Typically, traffic congestion can be expected during morning (7 AM-10 AM) and evening (5 PM-8 PM) hours.
- **Congestion-Prone Areas:** Traffic is often heavier in and around small towns with marketplaces or industrial zones.

4. Assessment of Road Quality and Infrastructure:

- **Road Quality:** Generally, roads are a mix of well-maintained highways and uneven rural roads. Potholes and narrow segments may pose challenges.
- **Infrastructure:** Bridges and culverts are generally in good condition, but some may have height or weight restrictions.

5. Suggestions for Alternative Routes:

In case of emergencies:

- **Alternative Routes:** Consider using nearby parallel state highways that connect through different rural pathways, although these may add to the travel time but can avoid congested towns or monsoon-affected areas.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- **Regulations:** Transport of hazardous materials is governed by strict guidelines, including night travel prohibitions, specific permits, and adherence to speed limits.
- **Checkpoints:** Regular inspection checkpoints are present, requiring documentation regarding the hazardous materials being transported.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- **Incidents:** Historical data indicate occasional accidents involving heavy vehicles primarily due to human error or challenging road conditions, particularly during adverse weather.

8. Environmental Considerations and Sensitive Areas:

- **Sensitive Areas:** The route might pass near agricultural lands and water bodies, requiring measures to prevent contamination or spill oversight.
- **Environmental Precautions:** Mitigation strategies for minimizing air and noise pollution are advisable.

9. Analysis of Communication Coverage:

- **Potential Dead Zones:** Cellular coverage can be spotty in rural segments and mountainous regions; emergency communications such as satellite phones or radios are recommended.

10. Estimated Emergency Response Times:

- **Response Times:** Typically, emergency response times may vary from 30 to 60 minutes in urban areas but could be longer in rural settings due to the distance from major cities.

11. Overall Summary of Risk Assessment:

- **Risk Level:** Moderate, due to the potential for weather-related disruptions, road quality variances, and communication challenges.
- **Recommendation:** Plan transit during daylight, ensure weather monitoring, and use designated rest areas for driver fatigue management. Familiarize the driver with local regulations regarding carrying hazardous materials and emergency routes.

This assessment aims to ensure safety and preparedness in mitigating risks on this specific transport route. Regular updates and checks are essential given the dynamic nature of road conditions and regulations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	Medium	11.45000, 77.87213	30 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.45309, 77.87325	30 KM/Hr
6	Turn	Medium	11.46400, 77.87323	30 KM/Hr
7	Turn	Medium	11.46765, 77.87037	30 KM/Hr
8	Turn	High	11.47403, 77.86841	15 KM/Hr
9	Turn	High	11.49149, 77.89601	15 KM/Hr
10	Turn	High	11.49224, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	Medium	11.53501, 77.95707	30 KM/Hr
12	Turn	Medium	11.53601, 77.95856	30 KM/Hr
13	Turn	Medium	11.64180, 78.11808	30 KM/Hr
14	Turn	Medium	11.64325, 78.11993	30 KM/Hr
15	Turn	Medium	11.69519, 78.10775	30 KM/Hr
16	Turn	Medium	11.75835, 78.05016	30 KM/Hr
17	Turn	Medium	11.87942, 78.08197	30 KM/Hr
18	Turn	Medium	11.92585, 78.06507	30 KM/Hr
19	Turn	Medium	11.95818, 78.04978	30 KM/Hr
20	Turn	Medium	11.95876, 78.04990	30 KM/Hr
21	Turn	Medium	11.96035, 78.04946	30 KM/Hr
22	Turn	Medium	11.96304, 78.05136	30 KM/Hr
23	Turn	Medium	11.96399, 78.05135	30 KM/Hr
24	Turn	Medium	11.96458, 78.05084	30 KM/Hr
25	Turn	Medium	11.96496, 78.04988	30 KM/Hr
26	Turn	High	11.96580, 78.04923	15 KM/Hr
27	Turn	Medium	11.96675, 78.04939	30 KM/Hr
28	Turn	Medium	11.96791, 78.05070	30 KM/Hr
29	Turn	Medium	12.12058, 78.13283	30 KM/Hr
30	Turn	High	12.14089, 78.12817	15 KM/Hr
31	Turn	High	12.14358, 78.10833	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
23	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
32	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
33	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
35	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
36	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
37	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
38	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
39	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
40	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
24	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
30	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
31	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
34	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
41	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69519, 78.10775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75835, 78.05016



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95876, 78.04990



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96035, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96458, 78.05084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96496, 78.04988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12058, 78.13283



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14089, 78.12817



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14358, 78.10833